

Name: _____

Date: _____

PROBABILITY, (BASIC PROBABILITY SINGLE EVENTS)

LO: I can calculate the probability of a single event using fractions.

Basic probability

Probability measures how likely something is to happen, from 0 (impossible) to 1 (certain). It equals favourable outcomes $\div$ total outcomes.

Key rule: $P(\text{event}) = \text{number of ways the event can happen} \div \text{total number of equally likely outcomes}$.

Quick examples

	$P(\text{heads on a coin}) = \frac{1}{2}$	1 out of 2 outcomes
	$P(6 \text{ on a die}) = \frac{1}{6}$	1 out of 6 outcomes
	$P(\text{not } A) = 1 - P(A)$	complementary events
	$P(\text{heads}) = \frac{1}{3}$ because heads, tails, or edge	edge is not equally likely
	$P(6 \text{ on a die}) = \frac{1}{3}$ because	

Key idea

Probability = favourable outcomes divided by total equally likely outcomes.

$$P(\text{event}) = \frac{\text{favourable}}{\text{total}}$$

Always a value between 0 and 1

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – single die

A fair six-sided die is rolled. Find $P(\text{even number})$.

Even numbers: 2, 4, 6 (3 outcomes)

Total outcomes: 6

$$P(\text{even}) = \frac{3}{6} = \frac{1}{2}$$

$$= \frac{1}{2}$$

Count the favourable outcomes and divide by total.

Example 2 – bag of sweets

A bag has 3 red, 5 blue, and 2 green sweets. Find $P(\text{blue})$.

Favourable (blue): 5

Total: $3 + 5 + 2 = 10$

$$P(\text{blue}) = \frac{5}{10} = \frac{1}{2}$$

$$= \frac{1}{2}$$

Add all sweets for the total, then simplify.

Example 3 – complement

$P(\text{rain}) = 0.3$. Find $P(\text{no rain})$.

$P(\text{no rain}) = 1 - P(\text{rain})$

$$= 1 - 0.3$$

$$= 0.7$$

$$= 0.7$$

Complementary probabilities always add up to 1.

Example 4 – spinner

A spinner has sections labelled 1 to 8. Find $P(\text{prime number})$.

Prime numbers 1-8: 2, 3, 5, 7 (4 outcomes)

Total: 8

$$P(\text{prime}) = \frac{4}{8} = \frac{1}{2}$$

$$= \frac{1}{2}$$

List the primes carefully (1 is not prime).

YOUR TURN – PRACTICE (deliberate practice)

1. Dice probabilities

A fair six-sided die is rolled. Find each probability.

- a) P(3)
- b) P(odd)
- c) P(greater than 4)
- d) P(less than 5)

2. Bag probabilities

A bag has 4 red, 6 blue, and 5 green balls. Find each probability.

- a) P(red)
- b) P(blue)
- c) P(green)
- d) P(not red)

3. Complementary probabilities

Use $P(\text{not } A) = 1 - P(A)$.

- a) P(win) = 0.6. Find P(not win)
- b) P(late) = $\frac{1}{5}$. Find P(not late)
- c) P(rain) = 0.35. Find P(no rain)
- d) P(pass) = $\frac{7}{10}$. Find P(fail)

4. Spinners and cards

Find each probability.

- a) A card from a pack of 52: P(heart)
- b) A card from a pack of 52: P(king)
- c) Spinner 1-10: P(multiple of 3)
- d) Spinner 1-10: P(square number)

5. Mixed single event probability

Calculate each probability as a simplified fraction.

- a) Letter from MATHEMATICS: P(vowel)
- b) Number 1-20: P(factor of 12)
- c) Bag: 7 red, 3 blue. P(blue) after adding 2 blue
- d) Die: P(prime or 1)
- e) $P(A) = x$. P(not A) in terms of x

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) P(event) is always between 0 and 1 ☐

Correct answer: _____

Reason: _____

b) If you flip a coin 10 times you will always get 5 heads ☐

Correct answer: _____

Reason: _____

c) $P(\text{not } A) = 1 - P(A)$ ☐

Correct answer: _____

Reason: _____

d) A probability of 0.5 means the event is certain ☐

Correct answer: _____

Reason: _____

e) P(6 on a fair die) = $\frac{1}{6}$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A bag contains red and blue counters. The probability of picking red is $\frac{3}{8}$. There are **9 red counters**. How many blue counters are there? If 3 blue counters are removed, what is the new P(red)?

Expression: _____

Simplified: _____

b) Two spinners each have sections 1-4. List all **16 possible outcomes** when both are spun. Find P(total = 5) and P(both the same number).

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

